Lecture 4: Applications

Computational cost

Elim. on $n \times n$ matrix.

$$\begin{bmatrix} 0 & & & & \\ & 0 & & & \\ & & \ddots & & \\ & & & 0 & \\ & & & & 0 \end{bmatrix}$$

$\hookrightarrow 2^2 - 2$

$\hookrightarrow (n-1)^2 - (n-1)$

$\hookrightarrow n(n-1) = n^2 - n$

operations $\left\{ \begin{array}{l} \text{multiplications followed by} \\ \text{subtraction} \end{array} \right.$

$$\rightsquigarrow (1^2 + 2^2 + \cdots + n^2) - (1 + 2 + \cdots + n)$$
$$= \frac{1}{6} n(n+1)(2n+1) - \frac{1}{2} n(n+1)$$
$$= \frac{1}{3}(n^3 - n) \approx \frac{n^3}{3} \text{ operations}$$

$n = 1000 \rightsquigarrow n^3 = (10^3)^3 = 10^9 = 1$ billion

ok for home PC/phone

$n = 1$ million $\rightsquigarrow n^3 = (10^6)^3 = 10^{18}$

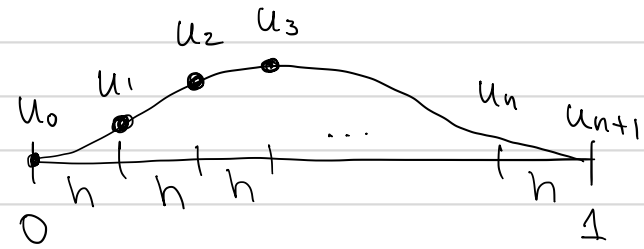
Not ok for home PC

(Iterative methods)

(1D) Poisson eqn: $-\dfrac{d^2u}{dx^2} = f(x)$

$$u(0) = u(1) = 0.$$

E.g.: u = temperature distribution in a rod, ends at $0°C$.



$$u_j := u(j \cdot h)$$

$$\frac{du}{dx} \approx \frac{\Delta u}{\Delta x} = \frac{u(x+h) - u(x)}{h}$$

or

$$\frac{u(x) - u(x-h)}{h}$$

or

$$\frac{u(x+h) - u(x-h)}{2h} \qquad (*)$$

$$\frac{d^2u}{dx^2} \approx \frac{\Delta^2 u}{\Delta x^2} = \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$-u_{j+1} + 2u_j - u_{j-1} = h^2 \cdot f(jh).$$

$$u_0 = u_{n+1} = 0.$$

$$-u_{j+1} + 2u_j - u_{j-1} = h^2 \cdot f(jh).$$

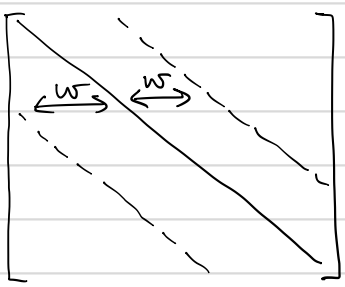
$$u_0 = u_{n+1} = 0.$$

$$\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \end{bmatrix} = \begin{bmatrix} f(h) \\ f(2h) \\ f(3h) \\ f(4h) \\ f(5h) \end{bmatrix}$$

Elimination:

$$\left[\begin{array}{ccccc|c} 2 & -1 & 0 & 0 & 0 & f_1 \\ 0 & \frac{3}{2} & -1 & 0 & 0 & f_2 + \frac{1}{2} f_1 \\ 0 & 0 & \frac{4}{3} & -1 & 0 & f_3 + \frac{2}{3}(*) \\ 0 & 0 & 0 & \frac{5}{4} & -1 & f_4 + \frac{3}{4}(**) \\ 0 & 0 & 0 & 0 & \frac{6}{5} & f_5 + \frac{4}{5}(***) \end{array}\right]$$

* Positive pivots, converge to 1.
* $\det A = 6$
* cost : $2n$ (Your PC can handle $n = 1$ billion!)



$\left(a_{ij} = 0 \text{ if } |i-j| \geq w \right)$

$\Rightarrow$ Elimination costs $\sim w^2 n$